

Day 4 Plan

Maths: Simple & Compound Interest · Reasoning: Classification (odd one out) ·
Science: Physics 4: Pressure, Fluids, Properties of Matter & Heat

Today's schedule

#	Block	Time	What you do	Where
0	Day 3 recall	15 min	10 questions from Day 3, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	Read the formulas & methods (10 min) → 2-3 book questions per pattern, timed → log mistakes	Section A
-	Break	10 min	Away from the screen	-
2	Reasoning	50 min	Learn the method (10 min) → 15 timed questions from your book → log	Section B
-	Break	10 min	Away from the screen	-
3	Science	60 min	Lessons with quick checks → 20-question practice set → answers	Section C
4	Current affairs	20 min	20 MCQs from the monthly PDF	Your PDF
5	Recall	10 min	Redo today's wrong questions; write 3 rules for tomorrow	Mistake log

Speed targets: Maths 60-70 sec per question · Reasoning 30-40 sec · Science 15-20 sec.

Block 0 — Day 3 recall (answers on the last page)

#	Question	Your answer
1	Work done when the force is at 90° to the motion?	
2	If speed doubles, KE becomes ...?	
3	Formula for potential energy?	
4	Energy conversion in an electric motor?	
5	1 HP = ? W	
6	Distance between two masses doubled → force becomes?	
7	Where is g maximum on Earth?	
8	Escape velocity from Earth?	
9	Period of a geostationary satellite?	
10	Maths: marked 40% above CP, 20% discount → profit %?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Simple & Compound Interest

Core idea

Simple interest (SI) is paid only on the original sum, so it's the **same every year**.

Compound interest (CI) adds each year's interest to the sum, so you earn **interest on interest** and it grows faster. For CI, use fractions instead of decimals: 10% = 1/10, so each year multiplies by **11/10**.

Formulas

Concept	Formula / rule
Simple interest	$SI = P \times R \times T / 100$ · Amount = P + SI
Compound amount	$A = P(1 + R/100)^n$ · CI = A – P
Half-yearly / quarterly	Rate R/2 , time 2n · quarterly: R/4, 4n
CI – SI, 2 years	$P(R/100)^2$
CI – SI, 3 years	$P(R/100)^2(3 + R/100)$
Effective rate, 2 years at R% CI	$2R + R^2/100$ % (10% → 21%)
SI: sum becomes n times	$R \times T = 100(n - 1)$ (doubles: RT = 100)
CI: x times in n years	becomes x^k times in kn years
Different rates each year	$A = P(1 + a/100)(1 + b/100)...$
Growth / depreciation	$P(1 + r/100)^n$ · $P(1 - r/100)^n$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Basic SI	PRT/100	₹5000 at 8% for 3 years → ₹1200
2. Find the rate or time	Rearrange: $R = SI \times 100 / (P \times T)$	SI ₹640 on ₹2000 for 4 years → 8%
3. Doubling / tripling (SI)	$RT = 100(n - 1)$	Doubles in 8 years → R = 12.5% ; to triple: $RT = 200$ → 16 years
4. Basic CI	$P(1 + R/100)^n$	₹10,000 at 10% for 2 years → $10,000 \times 121/100 = 12,100$ → CI ₹2100
5. CI – SI difference	$P(R/100)^2$	₹5000 at 10% for 2 years → $5000 \times 1/100 =$ ₹50
6. Half-yearly	R/2 and 2n	₹8000 at 10% p.a., half-yearly, 1 year → $8000 \times 1.05^2 =$ ₹8820
7. Effective rate	$2R + R^2/100$	10% CI for 2 years = 21% in total
8. CI multiples	x^k in kn years	Doubles in 5 years → 8 (= 2^3) times in 15 years
9. Different yearly rates	Multiply the factors	₹10,000 at 10% then 20% → $10,000 \times 1.1 \times 1.2 =$ ₹13,200
10. Depreciation	$P(1 - r/100)^n$	₹50,000 machine, 10% per year, 2 years → $50,000 \times 0.81 =$ ₹40,500

Traps

- SI interest is the **same every year**; CI interest **grows** every year.
- Time given in months → divide by 12. 'Compounded half-yearly' → halve the rate **and** double the periods.
- The question may ask for the **amount** or only the **interest** — read the last line again before marking.

B · REASONING — Classification (odd one out)

Three or four items share a property; one doesn't. **Method:** find the property that the **majority** share, then check each item against it. If you find two possible properties, prefer the **more specific** one (e.g. 'perfect squares' beats 'odd numbers').

Type	Properties to test (in this order)
Numbers	Prime · perfect square · perfect cube · even/odd · divisible by a common number · digit sum · $n^2 \pm 1$ · pairs following one rule
Letter groups	Gaps between letters (write the positions) · vowels · opposite pairs (sum 27) · reverse order · position sum
Words	Category (planets vs satellites, mammals vs fish, SI vs non-SI units, metals vs non-metals) · use · gender · synonyms

Patterns, with solved examples

Pattern	Example	Solution
Primes	3, 5, 7, 9, 11	9 is not prime → 9
Squares	16, 25, 36, 48, 64	48 is not a perfect square → 48
Cubes	27, 64, 125, 144, 216	$144 = 12^2$ is not a cube → 144
Number pairs	(3, 9), (4, 16), (5, 25), (6, 30)	Second = first ² except $6^2 = 36$ → (6, 30)
Letter gaps	BDF, HJL, MOQ, RTW	R, T, W = +2, +3 (others +2, +2) → RTW
Opposite pairs	AZ, BY, CX, DV	D's opposite is W → DV
Category	Mercury, Venus, Moon, Mars	The Moon is a satellite, not a planet → Moon
Units	Metre, kilogram, second, litre	Litre is not an SI base unit → litre
Animals	Whale, bat, shark, dolphin	Shark is a fish; the others are mammals → shark

Speed rule

Check the 'obvious' answer against all the other options before marking — examiners often add a second item that also looks odd at first glance.

C · SCIENCE — Physics 4: Pressure, Fluids, Properties of Matter & Heat

10 lessons — a big topic, so take 60 minutes. Part 1 (Lessons 1–7): pressure and fluids. Part 2 (Lessons 8–10): heat.

Lesson 1 · Pressure (about 4 min)

Pressure = force ÷ area ($P = F/A$), unit **pascal (Pa)** = 1 N/m². The same force on a **smaller area** gives **more pressure**.

Small area → large pressure (to cut or pierce)	Large area → small pressure (to avoid sinking or hurting)
Sharp knife · needle · nail tip · pointed teeth of animals	Wide straps on school bags · broad tyres on tractors and trucks · railway tracks laid on sleepers · camel's wide feet · snow shoes

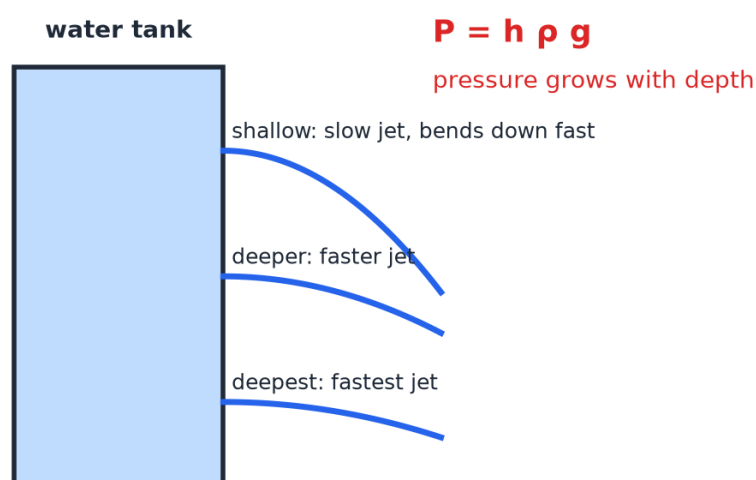
Quick check 1 — answer out loud before moving on (answers at the end)

Why are railway tracks laid on wide wooden or concrete sleepers?

Lesson 2 · Pressure in liquids (about 5 min)

A liquid presses on everything it touches — the base, the walls, and anything placed in it.

Liquid pressure $P = h \rho g$ (h = depth, ρ = density of the liquid, $g = 9.8$).



Property	Consequence
Increases with depth	Dam walls are made thicker at the bottom · divers feel pressure in their ears
Same at the same depth, and acts in all directions	Water squirts out of holes in all directions at the same depth
Does not depend on the shape or width of the vessel	Only depth and density matter
Greater in denser liquids	Mercury presses much harder than water at the same depth

Worked example — 10 m under water

$P = h \rho g = 10 \times 1000 \times 10 = 1 \times 10^5 \text{ Pa}$ — about the same as the whole atmosphere above us (1 atm).

Quick check 2 — answer out loud before moving on (answers at the end)

Why is a dam wall thicker at the bottom than at the top?

Lesson 3 · Atmospheric pressure (about 5 min)

Air has weight, so the atmosphere presses on us: **1 atm $\approx 1.013 \times 10^5 \text{ Pa} = 76 \text{ cm (760 mm) of mercury}$** . We don't feel it because the pressure inside our bodies balances it. It's measured with a **barometer**; a mercury column 76 cm high balances it (a water barometer would need about 10.3 m).

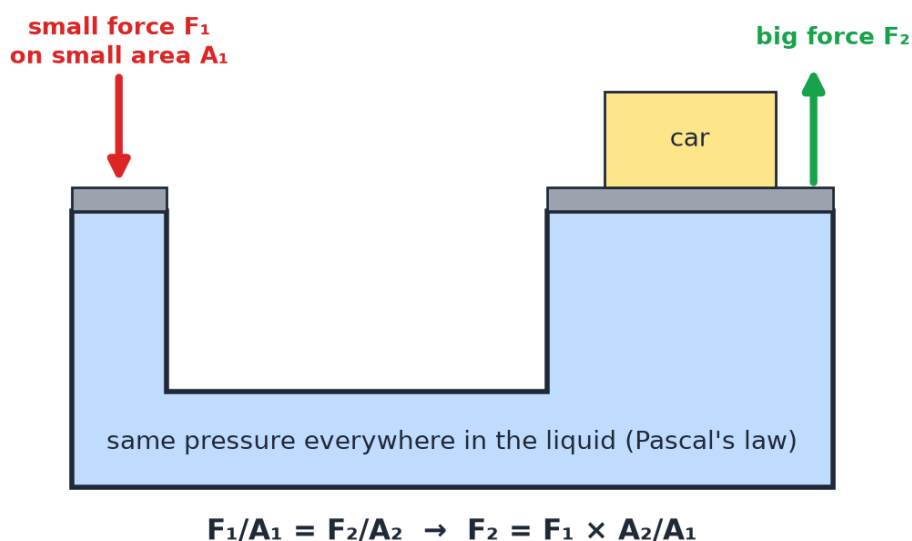
Fact	Explanation / example
Decreases with altitude	Nosebleeds at high altitude · a fountain pen may leak in an aeroplane · cooking is slow in the hills (water boils below 100 °C)
Things that work because of air pressure	Drinking straw · syringe · rubber sucker sticking to a wall · dropper
Barometer reading falls suddenly	Storm likely
Barometer reading falls gradually	Rain likely
Barometer reading rises steadily	Fair (dry) weather

Quick check 3 — answer out loud before moving on (answers at the end)

a) What is normal atmospheric pressure in cm of mercury? b) What does a sudden fall in barometer reading indicate?

Lesson 4 · Pascal's law and the hydraulic lift (about 4 min)

Pascal's law: pressure applied to a liquid in a closed container is **passed on equally** to every part of the liquid. Since $P = F/A$, a small push on a small piston produces a **huge force** on a large piston.



Worked example — hydraulic lift

Small piston area 0.01 m^2 , large piston 1 m^2 (100 times bigger). Push with 100 N on the small one.
 $F_2 = 100 \times (1 \div 0.01) = \mathbf{10,000 \text{ N}}$ — enough to lift a car.

Uses: **hydraulic lift** (car service stations), **hydraulic brakes** (vehicles), **hydraulic press**, JCB arms.

Quick check 4 — answer out loud before moving on (answers at the end)

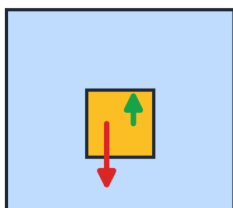
On what principle do hydraulic brakes work?

Lesson 5 • Buoyancy and Archimedes' principle (about 7 min)

A mug feels lighter under water. That's because liquids push objects **upward** — the **buoyant force** (upthrust). **Archimedes' principle:** a body immersed in a fluid feels an upthrust equal to the **weight of the fluid it displaces**.

red = weight (down) · green = upthrust/buoyant force (up)

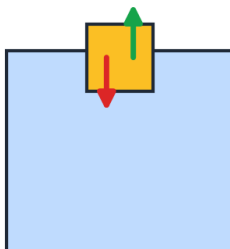
Weight > upthrust



SINKS

denser than the liquid (stone, iron nail)

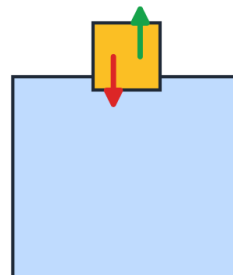
Weight = upthrust



FLOATS, mostly under

slightly less dense (ice: ~9/10 under)

Weight = upthrust



FLOATS high

much less dense (cork, thermocol)

Floating rule: an object floats if it's **less dense** than the liquid. A floating body sinks until it displaces **its own weight** of liquid (law of flotation).

Fact	Why
A steel ship floats, a steel nail sinks	The ship is hollow: it displaces a huge volume of water, enough to equal its weight
Ice floats with about 9/10 under water	Ice ($\approx 0.92 \text{ g/cm}^3$) is slightly less dense than water (1 g/cm^3)
Swimming is easier in sea water	Sea water is denser, so it gives more upthrust
A ship rises a little going from a river to the sea	Same reason: denser sea water $\rightarrow$ less volume needs to be displaced
Hydrometer / lactometer	Floating instruments that read density from how deep they sink

Worked example — relative density

An object weighs 10 N in air and 6 N in water. Upthrust = $10 - 6 = 4 \text{ N}$.

Relative density = weight in air $\div$ loss of weight in water = $10 \div 4 = \mathbf{2.5}$ (it's 2.5 times denser than water).

Quick check 5 — answer out loud before moving on (answers at the end)

a) Why does an iron ship float when an iron nail sinks? b) A body weighs 12 N in air and 8 N in water. What is its relative density?

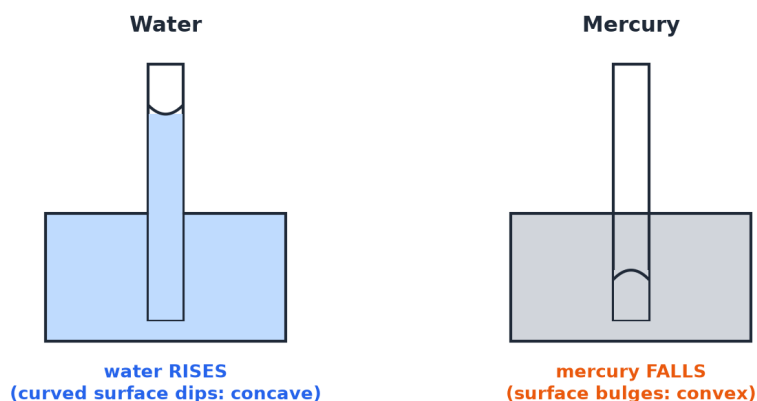
Lesson 6 · Surface tension, capillarity, viscosity and Bernoulli

(about 7 min)

Surface tension: a liquid's surface behaves like a **stretched elastic skin**, because surface molecules are pulled inward.

Effect of surface tension	Example
Drops take a spherical shape (least surface area)	Raindrops, mercury drops
Small insects can walk on water	Water striders
Soap and detergents reduce surface tension	Water spreads into fibres and cleans better
Surface tension decreases as temperature rises	Hot water cleans better than cold

Capillarity: a liquid rises (or falls) in a very narrow tube. Water **rises**; mercury **falls**. The narrower the tube, the higher the rise.



Examples: **oil or kerosene rising in a lamp wick**, ink absorbed by blotting paper, a towel soaking up water, water rising through tiny pores in soil and plants.

Viscosity is a liquid's internal friction — its resistance to flowing. Honey and glycerine are more viscous than water. Heating a **liquid lowers** its viscosity (warm honey flows easily); heating a **gas raises** it.

Bernoulli's principle: where a fluid moves **faster**, its pressure is **lower**. This explains the **lift on aeroplane wings**, roofs blown off during storms (fast wind above, still air below), two boats moving side by side being pulled together, and spray guns/atomisers.

Quick check 6 — answer out loud before moving on (answers at the end)

a) Why are raindrops spherical? b) Which principle explains the lift on an aeroplane wing?

Lesson 7 • Elasticity (about 2 min)

Elasticity is a material's ability to return to its original shape when the force is removed.

Hooke's law: within the elastic limit, **stress** $\propto$ **strain** (stress = force/area, strain = change in length/original length).

Exam trap

Steel is more elastic than rubber. In physics, 'more elastic' means it resists deformation more strongly and returns fully — steel has a much higher Young's modulus. This is a very common question.

Quick check 7 — answer out loud before moving on (answers at the end)

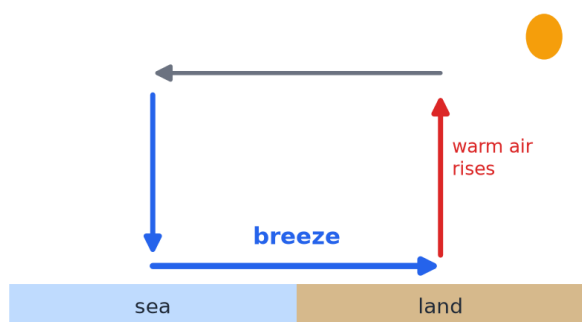
Which is more elastic: steel or rubber?

Lesson 8 • Heat transfer: conduction, convection, radiation (about 6 min)

Heat moves from hot to cold in three ways:

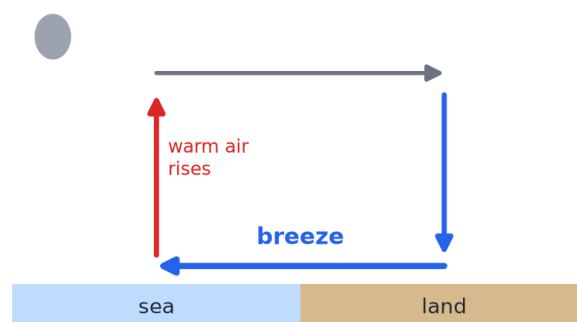
Mode	How	Examples
Conduction	Particle to particle, without the material moving — mainly in solids	A spoon's handle heats up in hot tea · metals are good conductors (silver best, then copper) · wood, plastic, air are poor (insulators) · woollen clothes trap air
Convection	The heated fluid itself moves (hot rises, cool sinks) — liquids and gases	Boiling water circulating · sea and land breezes · ventilators placed near the ceiling
Radiation	Electromagnetic waves — needs no medium	The Sun's heat reaching Earth through space · dark surfaces absorb and emit more · white or shiny surfaces reflect

DAY: sea breeze (sea → land)



land heats and cools faster than sea (water's high specific heat)

NIGHT: land breeze (land → sea)



land heats and cools faster than sea (water's high specific heat)

Thermos flask: the vacuum between the double walls stops **conduction and convection**; the silvered walls reduce **radiation**. Light-coloured clothes keep you cool in summer (reflect radiation); dark clothes keep you warm in winter.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Which mode of heat transfer needs no medium? b) When does a sea breeze blow — by day or by night?

Lesson 9 · Specific heat, latent heat, boiling and evaporation

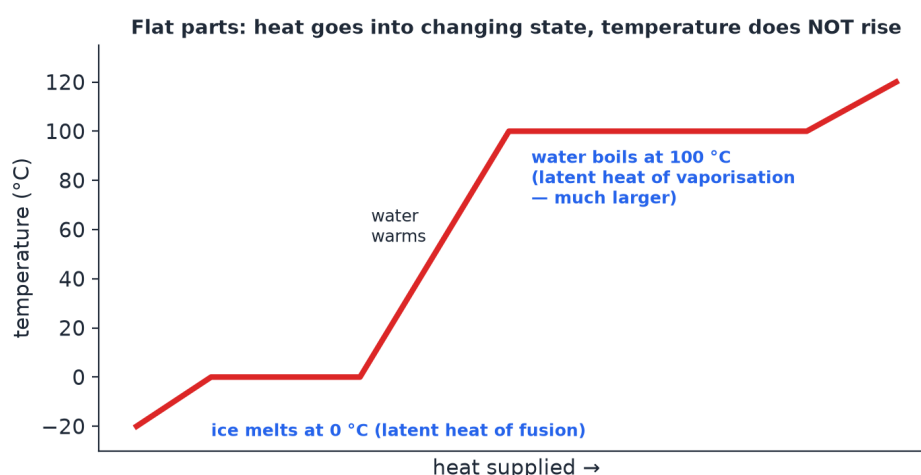
(about 7 min)

Specific heat (c) = heat needed to raise 1 kg of a substance by 1 °C. Heat needed: $Q = m c \Delta T$. **Water has a very high specific heat** ($\approx 4186 \text{ J/kg}^\circ\text{C}$): it heats and cools slowly. That's why coastal areas have a **moderate climate**, and why water is used in car radiators and hot-water bottles.

Worked example — $Q = mc\Delta T$

Heat to warm 2 kg of water by 10 °C = $2 \times 4186 \times 10 = 83,720 \text{ J} \approx \mathbf{84 \text{ kJ}}$.

Latent heat = heat absorbed or released during a **change of state** at constant temperature. During melting and boiling, the temperature stays the same even though heat is still going in:



Fact	Why
Steam at 100 °C burns worse than boiling water at 100 °C	Steam releases its large latent heat of vaporisation ($\approx 2260 \text{ kJ/kg}$) as it condenses on the skin
Ice at 0 °C cools a drink better than water at 0 °C	Ice absorbs latent heat of fusion ($\approx 334 \text{ kJ/kg}$) while melting
Food cooks faster in a pressure cooker	Higher pressure raises the boiling point ($\approx 120 \text{ }^\circ\text{C}$)
Cooking is slow in the hills	Lower pressure lowers the boiling point
Salt is spread on icy roads	Impurities lower the melting point of ice
Water in an earthen pot stays cool; sweating cools us	Evaporation causes cooling . It's faster with heat, wind, large surface area and low humidity

Quick check 9 — answer out loud before moving on (answers at the end)

a) Why does steam cause worse burns than boiling water? b) Why does food cook faster in a pressure cooker?

Lesson 10 · Thermal expansion and the strange behaviour of water (about 4 min)

Most substances **expand when heated**. Hence gaps are left between railway rails and in bridges, electric wires sag in summer, and a **bimetallic strip** (two metals that expand differently, e.g. brass and iron) bends when heated — used in **thermostats** and fire alarms.

Anomalous expansion of water: from 0 °C to 4 °C water **contracts** when heated; above 4 °C it expands like everything else. So **water is densest at 4 °C**.

Consequence	Why
Lakes freeze from the top; fish survive below	Water at 4 °C (densest) sinks to the bottom; ice (lighter) stays on top and insulates
Water pipes burst in very cold weather	Water expands when it freezes into ice

Quick check 10 — answer out loud before moving on (answers at the end)

At what temperature is water densest?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Pressure	$P = F/A$ · pascal · sharp knife vs wide bag straps, sleepers, camel feet
Liquid pressure	$P = h\rho g$ · increases with depth · acts in all directions · independent of vessel shape · dams thicker at the bottom
Atmosphere	1 atm = 1.013×10^5 Pa = 76 cm Hg · falls with height · straw, syringe · sudden barometer fall = storm
Pascal	Pressure passed on equally · hydraulic lift, brakes, press · $F_2 = F_1 \times A_2/A_1$
Archimedes	Upthrust = weight of liquid displaced · floats if less dense · ship hollow · ice 9/10 under · sea water denser · $RD = W \div$ loss of weight
Surface tension	Spherical drops · insects on water · soap lowers it · falls with temperature
Capillarity	Water rises, mercury falls · lamp wick, blotting paper · narrower tube → higher rise
Viscosity / Bernoulli	Liquid viscosity falls with heat, gas rises · faster flow → lower pressure: aeroplane lift, roofs blown off
Elasticity	Hooke: stress $\propto$ strain · steel more elastic than rubber
Heat transfer	Conduction (solids, silver best) · convection (fluids, breezes) · radiation (no medium) · thermos: vacuum + silvering
Specific heat	$Q = mc\Delta T$ · water highest → moderate coastal climate, radiators
Latent heat	Steam burns worse · ice cools better · pressure cooker raises BP · hills lower BP · salt lowers MP · evaporation cools
Expansion	Rail gaps, bimetallic strip/thermostat · water densest at 4 °C · lakes freeze from the top · pipes burst

Science practice set — 20 questions · target 6 minutes

#	Question
1	The SI unit of pressure is: (a) newton (b) joule (c) watt (d) pascal
2	A sharp knife cuts better than a blunt one because it: (a) applies more pressure on a smaller area (b) applies more force (c) has less friction (d) is heavier
3	Pressure exerted by a liquid depends on its: (a) shape of the vessel (b) depth (c) area of the base (d) total volume
4	Dams are made thicker at the bottom because: (a) it looks better (b) liquid pressure increases with depth (c) water is lighter at the bottom (d) the soil is soft
5	A hydraulic lift works on: (a) Archimedes' principle (b) Bernoulli's principle (c) Pascal's law (d) Hooke's law
6	A fountain pen may leak in an aeroplane at high altitude because: (a) air pressure outside decreases (b) air pressure outside increases (c) the ink expands with cold (d) gravity decreases
7	Normal atmospheric pressure is about: (a) 76 mm of mercury (b) 76 cm of mercury (c) 10 cm of mercury (d) 760 cm of mercury
8	A sudden fall in barometer reading indicates: (a) fair weather (b) snowfall (c) a storm (d) an earthquake
9	A ship rises a little when it moves from a river into the sea because: (a) sea water is denser (b) sea water is lighter (c) the sea is deeper (d) of tides
10	An object weighs 10 N in air and 6 N in water. Its relative density is: (a) 1.67 (b) 2.5 (c) 4 (d) 0.6
11	Raindrops are spherical because of: (a) surface tension (b) viscosity (c) gravity (d) air pressure
12	Kerosene rises in the wick of a lamp due to: (a) surface tension only (b) capillarity (c) viscosity (d) buoyancy
13	The lift on an aeroplane wing is explained by: (a) Bernoulli's principle (b) Pascal's law (c) Archimedes' principle (d) Newton's first law
14	Which of these is the most elastic? (a) rubber (b) steel (c) plastic (d) clay
15	A sea breeze blows: (a) from land to sea during the day (b) from sea to land during the day (c) from sea to land at night (d) only during storms
16	The vacuum between the walls of a thermos flask reduces heat loss by: (a) conduction and convection (b) radiation only (c) evaporation (d) reflection
17	Steam at 100 °C causes more severe burns than water at 100 °C because steam: (a) is hotter (b) has higher specific heat (c) has extra latent heat of vaporisation (d) moves faster
18	Food cooks faster in a pressure cooker because: (a) the boiling point of water falls (b) steam is trapped and cools (c) less heat is lost by radiation (d) the boiling point of water rises
19	Water has its maximum density at: (a) 0 °C (b) 100 °C (c) 4 °C (d) -4 °C
20	Coastal areas have a moderate climate because water has a high: (a) specific heat (b) density (c) viscosity (d) surface tension

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	d	Lesson 1
2	a	Lesson 1
3	b	Lesson 2
4	b	Lesson 2
5	c	Lesson 4
6	a	Lesson 3
7	b	Lesson 3
8	c	Lesson 3
9	a	Lesson 5
10	b	$10 \div (10 - 6)$ (Lesson 5)
11	a	Lesson 6
12	b	Lesson 6
13	a	Lesson 6
14	b	Lesson 7
15	b	Lesson 8
16	a	Lesson 8
17	c	Lesson 9
18	d	Lesson 9
19	c	Lesson 10
20	a	Lesson 9

Quick checks

QC	Answers
1	Sleepers spread the train's weight over a large area, reducing the pressure on the ground
2	Liquid pressure increases with depth, so the bottom must withstand the most pressure
3	a) 76 cm b) a storm
4	Pascal's law
5	a) The hollow ship displaces enough water to equal its weight; the nail is denser than water b) $12 \div (12 - 8) = 3$
6	a) Surface tension pulls drops into the shape with least surface area b) Bernoulli's principle
7	Steel
8	a) Radiation b) By day (land heats faster, warm air rises over land, cool air flows in from the sea)
9	a) Steam releases latent heat of vaporisation when it condenses b) Higher pressure raises the boiling point of water
10	4 °C

Day 3 recall (Block 0)

#	Answer	#	Answer
1	Zero	6	$\frac{1}{4}$
2	4 times	7	At the poles
3	mgh	8	11.2 km/s
4	Electrical → mechanical	9	24 hours
5	746 W	10	$40 - 20 - 8 = 12\%$

Send me tonight

Day 3 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.